

BB84

Turning no-cloning into a key exchange

Lecture 19

Term 261

Where we are

Everything so far has been about *preserving* quantum information. Quantum key distribution turns the same physics around: the fact that measurement disturbs a state becomes a feature, because it makes eavesdropping detectable.

What QKD actually provides

Not encryption. QKD produces a **shared random secret key** between two parties, with a bound on how much an eavesdropper could know about it. The key is then used with a classical cipher — ideally the one-time pad, in practice AES.

The security argument, in one line: an eavesdropper cannot copy an unknown quantum state (no-cloning, L11), so any information she gains must disturb the transmission, and that disturbance shows up as errors Alice and Bob can measure.

History of QKD

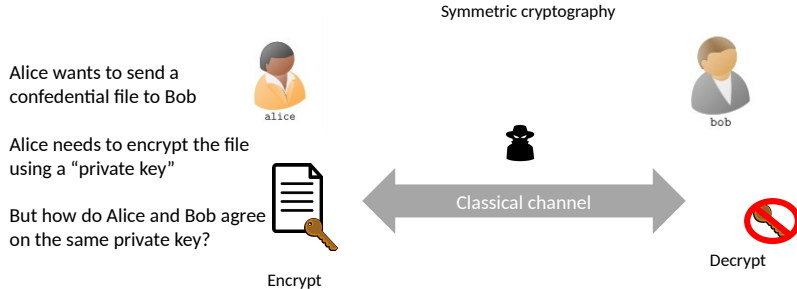
- Stephen Wiesner – early 1970s wrote paper "Conjugate Coding"
- Paper by Charles Bennett and Gilles Brassard in 1984 is the basis for QKD protocol BB84
- Another QKD protocol was invented independently by Artur Ekert in 1991 (E91)
- Later protocols include Device Independent QKD and Twin-Field QKD to remove the assumption of trusted hardware

Two Protocols for QKD

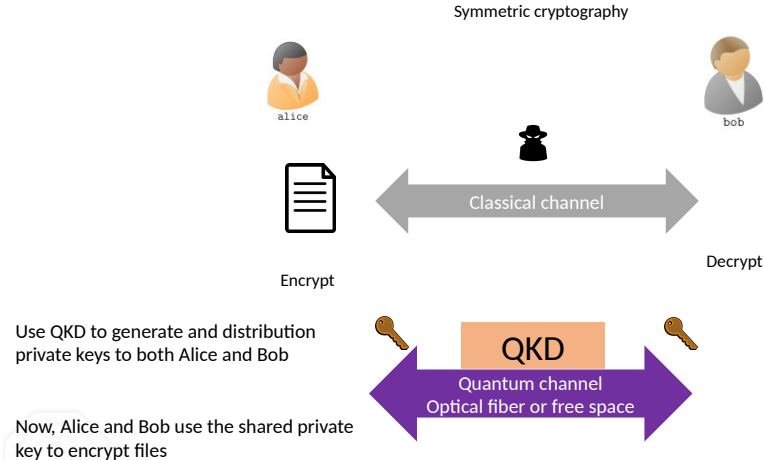
- BB84
 - Uses polarization of photons to encode the bits of information
 - Relies on “uncertainty” to keep Eve from learning the secret key.
- E91
 - Uses entangled photon states to encode the bits
 - Relies on the fact that the information defining the key only "comes into being" after measurements performed by Alice and Bob

■ Bennett: “Quantum cryptography using any two nonorthogonal states”, *Physical Review Letters*, Vol. 68, No. 21, 25 May 1992, pp 3121-3124 Ekert, Artur K. "Quantum cryptography based on Bell's theorem." *Physical review letters* 67, no. 6 (1991): 661.

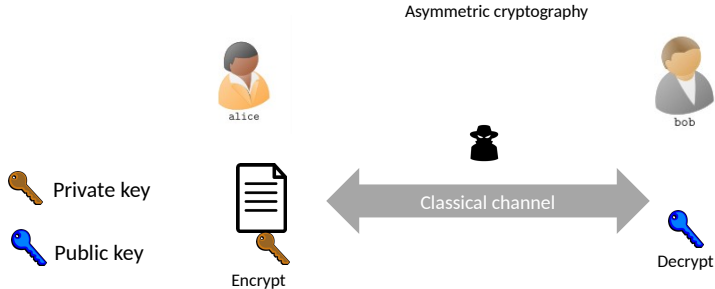
Basic Idea - Symmetric Cryptography



Basic Idea - Symmetric Cryptography with QKD



Basic Idea – Asymmetric Cryptography



Most common algorithm to generate public-private key is RSA

Shor's algorithm is proven to break RSA

Solution -> Post-Quantum Cryptography (more into this in ICS562)



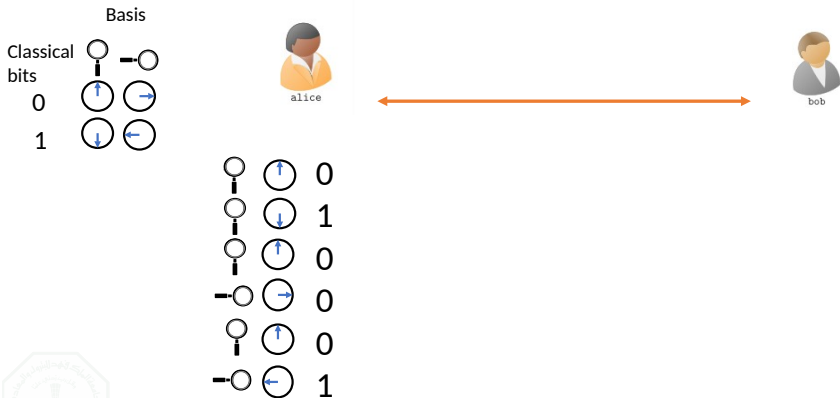
BB84 – Step by step protocol

- Alice and Bob agree on the basis for generating and measuring qubits



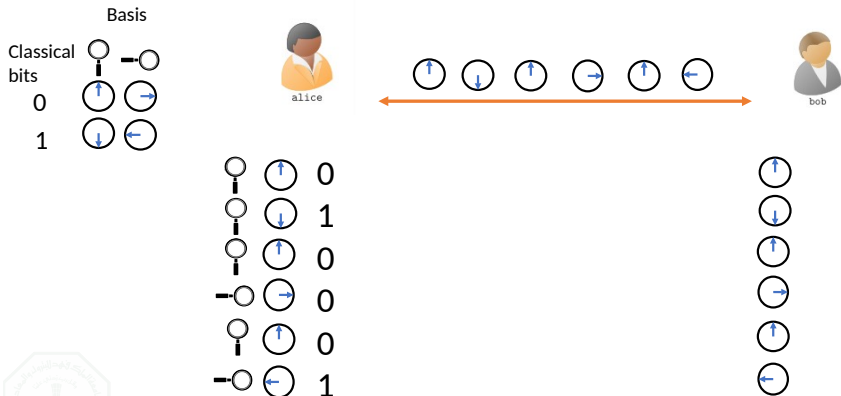
BB84 – Step by step protocol

- Alice chooses the basis and generates random classical bits accordingly



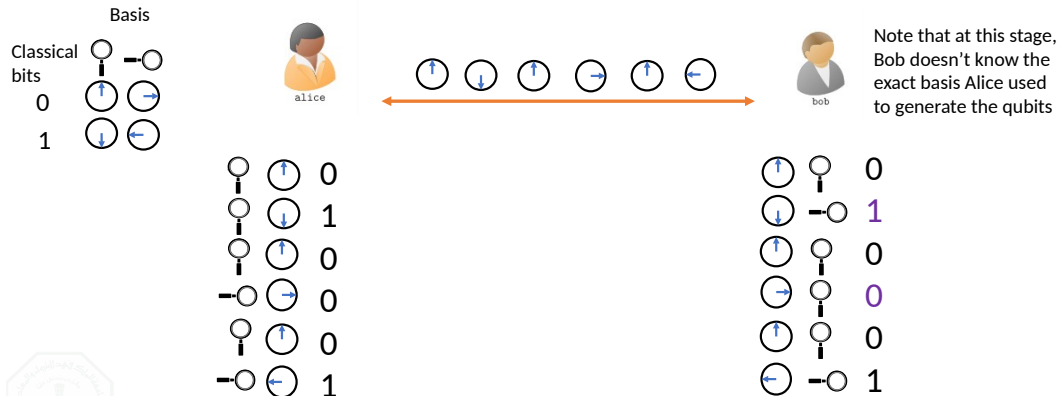
BB84 – Step by step protocol

- Alice transmits the polarized qubits using quantum channels to Bob



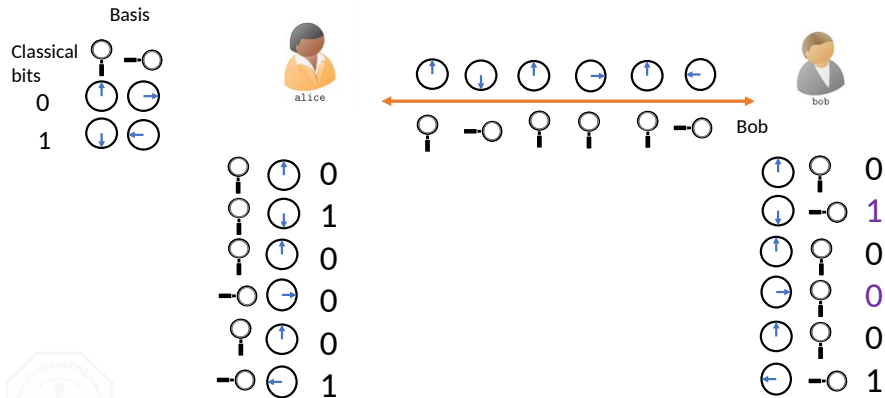
BB84 – Step by step protocol

- Bob measures the qubits in a random sequence of bases



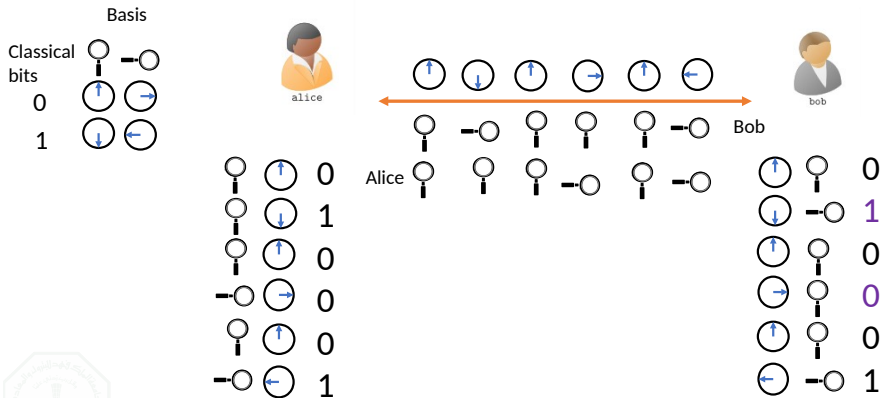
BB84 – Step by step protocol

- Bob tells the sender publicly what sequence of bases were used



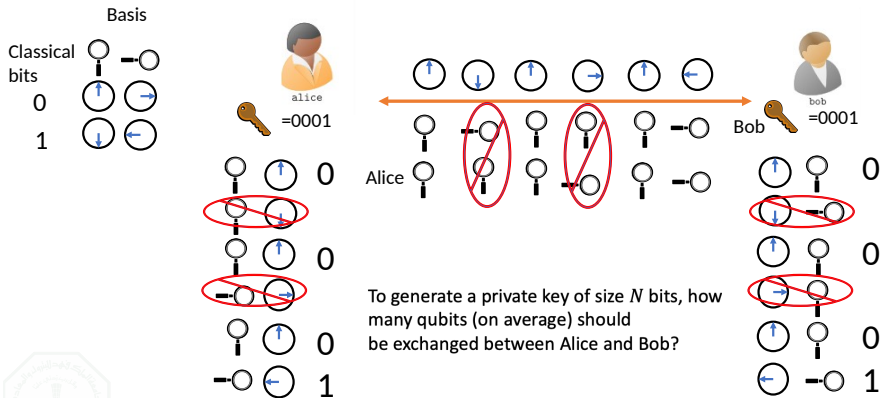
BB84 – Step by step protocol

- Alice tells the receiver publicly which bases were correctly chosen.



BB84 – Step by step protocol

- Alice tells the receiver publicly which bases were correctly chosen.



Sifting, and the numbers that come out

After transmission Alice and Bob announce their *bases* — never their bits — and keep only the positions where the bases matched.

- Bases agree half the time on average, so the sifted string is about half the length
- Where the bases agree and there is no noise or eavesdropper, the bits agree perfectly
- Where the bases disagree, Bob's outcome is uniformly random and carries nothing

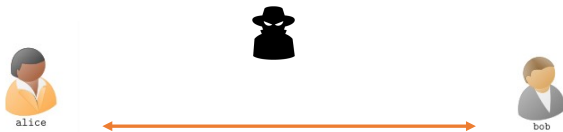
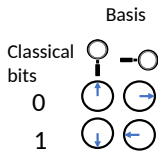
Stage	Fraction	Why
Sent	1	—
Detected	η	channel loss
Sifted	$\eta/2$	bases mismatch
After error correction	less	leakage to Eve
After privacy amplification	less still	Eve's knowledge

Then estimate the error rate. Alice and Bob sacrifice a random subset of the sifted bits and compare them publicly. The fraction that disagree is the **quantum bit error rate**, Q .

Q is the single number the security of the whole protocol rests on — and Alice and Bob must, conservatively, attribute *all* of it to Eve: rather than to their own imperfect hardware.

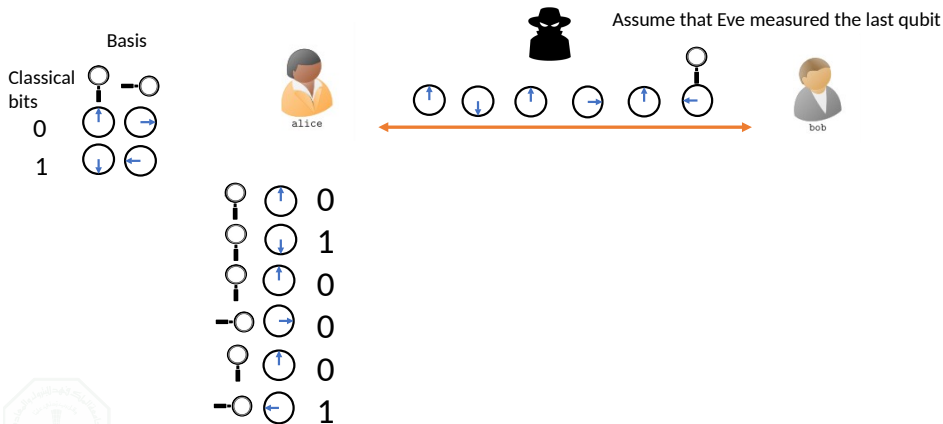
BB84 with Eavesdropping – Step by step protocol

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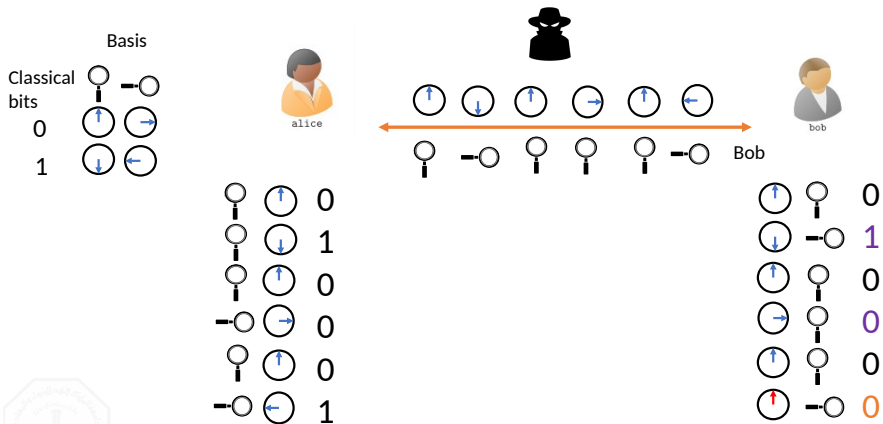
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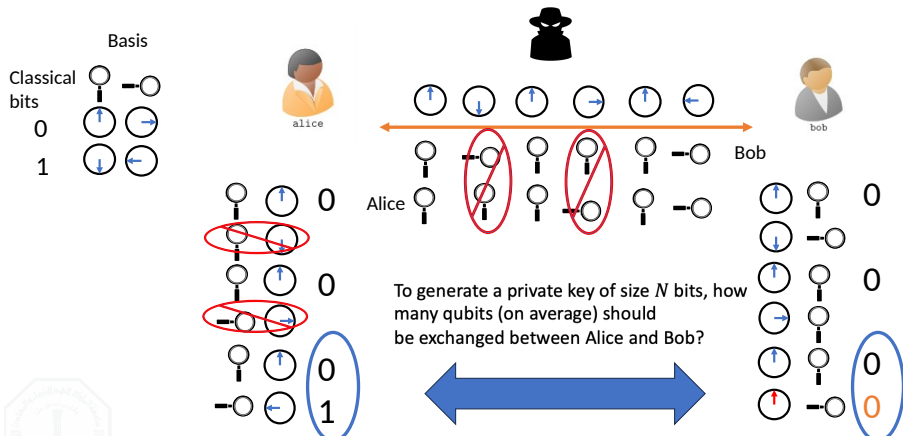
BB84 with Eavesdropping – Step by step protocol

- Bob tells the sender publicly what sequence of bases were used



BB84 with Eavesdropping – Step by step protocol

- Alice shared half of her bits with Bob to check for eavesdropping



Intercept–resend, quantified

The simplest attack: Eve measures each qubit in a randomly chosen basis and sends on what she found.

The arithmetic

- Eve picks the right basis half the time; those qubits pass through undisturbed
- The other half she measures in the wrong basis, and resends a state that is random in Alice's basis
- Of those, Bob gets the wrong bit half the time

$$Q = \frac{1}{2} \times \frac{1}{2} = 25\%.$$

What that tells Alice and Bob

- A full intercept–resend attack is unmissable — a quarter of the sifted bits disagree
- A partial attack on a fraction f of pulses gives $Q = f/4$, so the disturbance scales with how greedy Eve is
- Real systems see $Q \approx 1\text{--}5\%$ from hardware alone, so the interesting question is how much key survives at a given Q

| *Intercept–resend costs Eve 25% error for partial information. Is there a smarter attack that gains the same information for less disturbance — and if so, what would set the limit on how good it can be?*

Takeaway questions

- 1 State precisely what QKD delivers and what it does not. Why is it not an encryption scheme?
- 2 Walk through BB84 and say at each step what is public and what stays secret.
- 3 Derive the 25% error rate for a full intercept–resend attack.
- 4 Why must Alice and Bob attribute all observed errors to Eve, even when they know their detectors are imperfect?
- 5 Sifting discards about half the detections. What would be lost if Alice and Bob announced bits rather than bases?

Next

Alice and Bob now hold two *almost* identical strings, partially known to Eve.

- **L20** — reconciling them and squeezing out Eve's knowledge; the secret-key rate and the 11% threshold; then entanglement-based QKD
- **L21** — how real systems are attacked, and the lab experiment